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Set No:1

Code No: V3125/R07

III B.Tech I Semester Regular & Supplementary Examinations, November 2011

COMPUTER GRAPHICS

(Common to Computer Science and Engineering & Information Technology & Electronics and Computer Engineering)

Time: 3 Hours **Max Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

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1. a)Define computer graphics. List and explain the applications of computer graphics.
   b) Briefly explain about raster scan systems.

2. a) What is DDA? Digitize the line with endpoints (20,10) and (30,18) by using Bresenham's line drawing algorithm.
   b) Write a description of the midpoint circle algorithm in which decision parameter P is updated using x_{i+1} & y_{i+1} instead of x_i & y_i.

3. a) How are pictures made to grow, shrink and rotate? Explain with transformation matrices and examples
   b) Write short notes on transformations between coordinate systems.

4. a) Discuss about window to view port coordinate transformations in detail.
   b) Explain about Cyrus-Beck line clipping algorithm in detail.

5. a) How are periodic B-spline curves different from non-periodic B-spline curves?
   b) Explain the importance of B-Spline curves in geometric modeling

6. a) Explain various coordinate system in 3d viewing.
   b) Describe 3-D rotation about x, y and z axes and write the corresponding transformation matrices.

7. a) Discuss the techniques for efficient visible surfaces algorithm with suitable example.
   b) Briefly explain depth sorting in visible surface detection.

8. a) Discuss about a design of animation sequence. How frame-by-frame animation works?
   b) Briefly explain any four Animation techniques.

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Set No:2

Code No: V3125/R07

III B.Tech I Semester Regular & Supplementary Examinations, November 2011

COMPUTER GRAPHICS

(Common to Computer Science and Engineering & Information Technology & Electronics and Computer Engineering)

Time: 3 Hours **Max Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) How the interactive graphics display works? Explain them in detail with neat diagram and their working principles
   b) What is aliasing? Explain how one can reduce effect of aliasing.

2. Explain about boundary fill and flood fill algorithm in detail with suitable examples.

3. a) Define transformation. Explain in detail about basic 2-D geometrical transformations
   b) Write the general form of the matrix for rotation about a point P (h,k).

4. a) What happens to pictures that are too large to fit on the screen? Give various categories of methods in it
   b) Discuss in detail about Sutherland-Hodgman polygon clipping.

5. a) What is Polygon? Explain different type of polygons. Explain the various approaches used to represent polygon
   b) Explain the procedure for generation of various surfaces and curves using B-Spline method

6. a) Write the transformation matrix to rotate a point (x,y,z) about z-axis through an angle θ in clock wise direction.
   b) Discuss about combined 3d transformations

7. a) Explain the binary space partition tree visible surface Algorithm and write a pseudo code for building a BSP tree
   b) Give the advantages and limitations of various visible surface detection methods.

8. a) Discuss about general computer animation functions.
   b) How do you implement Morphing animation technique? Discuss with an example

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Set No:3

Code No: V3125/R07

III B.Tech I Semester Regular & Supplementary Examinations, November 2011

COMPUTER GRAPHICS

(Common to Computer Science and Engineering & Information Technology & Electronics and Computer Engineering)

Time: 3 Hours **Max Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) Explain the working principle of CRT with neat diagram.
   b) With the help of a diagram explain the architecture of Raster displays.

2. a) Describe briefly Bresenham's circle drawing algorithm with an example. Why do we prefer incremental algorithm over DDA?
   b) Explain Flood Fill algorithm for polygon filling with a suitable example

3. a) Write the homogeneous coordinate transformation matrices for the three basic transformations.
   b) Find reflections of point (3, 4) with respect to a line y = x and find the corresponding transformation matrix.

4. a) How do we represent polygon using polygon table, edge table and vertex table? Explain with an example.
   b) Distinguish between window and viewport. In 2-D clipping, how lines are grouped into visible, invisible and partially visible categories

5. a) How are periodic B-spline curves different from non-periodic B-spline curves?
   b) Given control points (10,100), (50,100), (70,120) and (100,150). Calculate coordinates of any four points lying on the corresponding Beizer curve.

6. a) Discuss about reflection and shear transformations.
   b) Give a 3x3 homogeneous matrix to rotate the image clockwise by 90^0. Then shift the image to the right by 10 units. Finally scale the image by twice as large.

7. a) Briefly explain area sub-division algorithm and give its advantages over other methods.
   b) What is ray tracing algorithm for hidden surface removal? Explain mathematically how do we find which planes are visible using ray tracing algorithm

8. a) Briefly explain the steps involved in the Design of animation sequence
   b) Define the term morphing and explain its use in key frame systems of animation

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Set No:4

Code No: V3125/R07

III B.Tech I Semester Regular & Supplementary Examinations, November 2011

COMPUTER GRAPHICS

(Common to Computer Science and Engineering & Information Technology & Electronics and Computer Engineering)

Time: 3 Hours **Max Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) Describe random and raster display systems with the help of neat sketches.
   b) Distinguish between aliasing and antialiasing.

2. a) Explain midpoint circle algorithm. Given a circle radius r=10, demonstrate the midpoint circle algorithm by determining positions along the circle octant in the first quadrant from x=10 to x=y.
   b) Explain DDA line drawing algorithm between points (a, b) and (c, d).

3. a) Write a note on window-to-viewport transformation.
   b) Determine sequence of basic transformations that are equivalent to the rotation of a point (x, y) around (1, 3).

4. a) Explain the concept of polygon clipping with neat diagrams.
   b) Clip a line segment between points (1, 3) to (5, 17) using Cohen Sutherland clipping algorithm so that it fit into view port with left bottom at (2, 5) and right top at (5, 12).

5. Determine the Bezier blending functions for five control points. Plot each function and label the maximum and minimum values.

6. a) Briefly explain about composite transformations.
   b) The triangle position is A(1,1),B(3,5) and C(2,2), translate the given triangle with respect to its centroid with A in the centroid position and rotate it by an angle 30^0.after shrinking the object half the size.

7. a) Briefly explain back-face detection algorithm.
   b) What are the two spaces in which hidden surface algorithm work? How does sorting and coherence speed up calculation in such algorithms.

8. a) Write short notes on general computer animation functions
   b) Describe in brief the steps required to produce real time animation.

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